Observable relaxation and perturbation detection near a fermionic Liouvillian exceptional point

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We derive exact observable selection rules for conditional relaxation near a fermionic Liouvillian exceptional point. With uniform multiparticle pumping, total particle number and a two-mode occupation imbalance probe different decay channels from the same strictly positive initial state. Exchange symmetry separates the signals, while fermion parity further restricts their accessible modes. A semiclassical analysis predicts a systematic separation of the two rates as the pump order increases. When imbalance decays more slowly, unequal detector weights expose its tail, potentially at a small survival probability. For discrimination of known pump strengths with a specified system-or-erasure output, short vacuum trials instead attain the optimal preparation-cost scaling, counting every failure. The exact signal reduction and matching detection bounds distinguish the spectral channels governing density relaxation from the resources governing pump discrimination.

Introduction. Exceptional points (EPs) are spectral degeneracies at which eigenmodes coalesce, producing nonexponential dynamics and unusual responses to perturbations [1, 2]. Connecting these spectral features to a density measurement requires identifying the modes reached by both preparation and readout. In open systems, quantum jumps distinguish Hamiltonian and Liouvillian EPs [3], and different correlation orders can reveal different decay structures [4]. The observable relaxation time is therefore a property of the selected signal as well as the spectrum.

Symmetries organize Liouvillian decay channels [5, 6] and allow observables to relax at different rates [7]. Preparation and readout can cancel Jordan contributions [8], as also occurs in response functions [9]; fermion parity further restricts one-time physical signals [10, 11]. Spectral methods resolve the sectors of quadratic open fermionic systems [12], and no-jump examples exhibit several coalescing mode families [13]. To assign a decay rate to a density signal, these sector constraints must be combined with its actual preparation and observable overlaps.

Perturbation detection also depends on which output and resources are counted. EP sensing includes resonant and amplifying schemes [14, 15], with precision determined by noise and resource constraints as well as response [16, 17]. Quantum Fisher information characterizes local estimation [18, 19], and postselection requires counting successful and failed trials [20–22]. Discriminating two specified alternatives defines an operational detection task [23]. Recent work addresses gapped Liouvillian EPs with access to the system and emitted fields [24]; here each trial outputs the retained system or a failure flag.

We derive the density-channel selection and detection cost in the uniformly pumped fermionic model of Xu and Yi [25]. At zero pump, its occupation decay modes coalesce at a high-order EP. A d-particle pump induces fractional spectral splitting with the established exponent $1/(d+1)$, consistent with earlier structured-EP results [26, 27]. Our exact reduction shows how two density observables select different relaxation channels from the same physical state. A short-time protocol and matching information bound then determine the preparation-cost scaling for pump discrimination, separating that task from observing a low-yield conditional tail. The Supplemental Material (SM) [28] gives the reductions and proofs.

Monitored dynamics and a common preparation. Consider $n \geq 2$ fermionic modes with number-conserving quadratic Hamiltonian $H = \sum_{ij} h_{ij} c_i^\dagger c_j$, $h = h^\dagger$. Each occupied mode can lose a particle at rate γ, while every empty d-mode subset is pumped with equal weight. The retained, unnormalized state evolves as

$$\dot{\widetilde{\rho}} = -\mathrm{i}[H, \widetilde{\rho}] + \gamma \sum_j c_j \widetilde{\rho} c_j^\dagger - n\gamma \widetilde{\rho} + z\gamma \sum_{|I|=d} c_I^\dagger \widetilde{\rho} c_I. \quad (1)$$

Here $1 \leq d \leq n$, and $c_I^\dagger$ creates one fermion in each mode of I. This is the recycling-only pump of Ref. [25]; adding the usual pump anticommutators would define a different model. The physical monitored realization requires $0 \leq z \leq n/\binom{n}{d}$. Within this range, additional discarded jumps complete the dynamics and are recorded as failures [29]. The trace of $\widetilde{\rho}$ is the survival probability P_{surv}; dividing by it gives the state conditioned on retention. SM Sec. I specifies the monitoring completion.

We measure the mean filling $\langle \hat{N} \rangle / n$ and the imbalance $\langle D \rangle$, where $\hat{N} = \sum_j n_j$ and $D = n_1 - n_2$. The occupations n_j refer to energy eigenmodes. Uniform pumping and loss permit this choice without changing their form. To prepare both signals, begin with independent mode occupations of probability p. Multiply the probabilities of the marked configurations 10 and 01 by $1+\epsilon$ and $1-\epsilon$, respectively, leaving 00 and 11 unchanged. For $0 < p < 1$ and $0 < |\epsilon| < 1$, this gives a normalized, strictly positive state with mean number np and imbalance $2\epsilon p(1-p)$. It commutes with fermion parity, although it populates both even and odd particle numbers.

Two observable relaxation channels. Total number and occupation imbalance select two different, exactly closed channels of the occupation dynamics. Total number probes the fully symmetric channel, while imbalance probes a channel with two fewer active occupation degrees of freedom. This separation follows from exchange

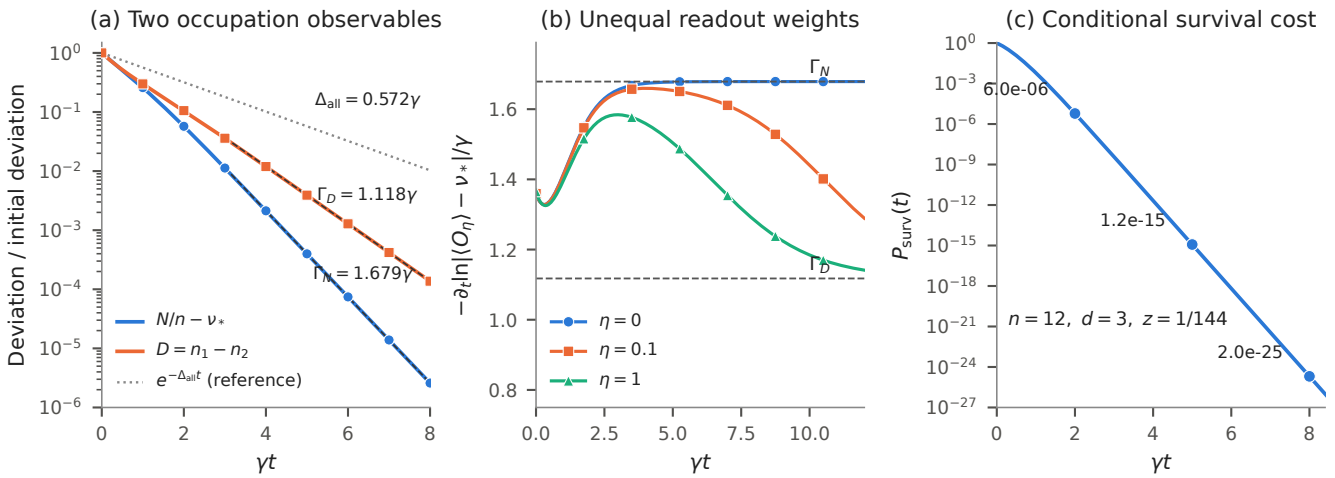


FIG. 1. Two relaxation rates from one preparation: $p = 0.75$, $\epsilon = 0.6$, $n = 12$, $d = 3$, and $z = 1/144$. (a) Conditional particle-number and imbalance deviations, normalized by their initial values; $\nu_* = \lim_{t\to\infty}\langle \hat{N}\rangle_t/n$ is the limiting mean filling. Dashed lines use exact spectral amplitudes and rates without fitting; the dotted unrestricted-gap reference describes neither signal. (b) Effective logarithmic decay rate of $O_\eta = \hat{N}/n + \eta D/n$ for different imbalance weights η. Horizontal lines mark Γ_N and Γ_D. (c) Survival probability of the same preparation.

symmetry: total number treats the marked modes identically, whereas imbalance changes sign when they are exchanged. The Hamiltonian leaves diagonal energy-basis populations unchanged, and uniform loss and pumping preserve their exchange symmetry. The two readouts therefore cancel opposite symmetry contributions while sharing the same survival normalization. Increasing ϵ changes only the imbalance amplitude, leaving the entire particle-number and survival curves unchanged. SM Sec. II derives the exact projection formulas.

The same preparation consequently displays two density-relaxation rates [Fig. 1(a)]. For $n = 12$ and three-particle pumping, the mean filling approaches its limiting value at $\Gamma_N = 1.678630\gamma$, while the imbalance vanishes at $\Gamma_D = 1.117616\gamma$. The unrestricted spectral gap, defined by the real-part separation from the leading mode, is $\Delta_{\rm all} = 0.571963\gamma$, below both observed rates. Its slower mode lies in the odd operator sector, inaccessible to a parity-even state and a parity-even one-time observable. The even-sector gap equals Γ_D in this example; total number excludes that mode by exchange symmetry. Fermionic two-time correlators can probe odd intermediate sectors and are outside this one-time restriction.

Within an allowed sector, the visible rate is selected by nonzero overlap with both the preparation and the normalized observable. Complex modes can produce oscillations, and coalescing modes can carry time-polynomial factors. At the physical small-system point $n = d = 4$, $z = 1$, oscillatory modes set the even-sector gap while the imbalance has a different exponential factor. This counterexample makes the overlap condition essential to any identification of a measured rate with a sector gap. SM Secs. II–III give the exact rate conditions and complete spectrum.

Changing detector weights can change the asymptotic rate with the generator and preparation held fixed. Unequal weights on the marked modes add an imbalance component to the number readout. Whenever this component is nonzero and decays more slowly, it eventually controls the measured tail [Fig. 1(b)]. The familiar competition of two decays is quantitative here because both rates and amplitudes are known. For imbalance weight $\eta = 0.1$, the asymptotic contributions become comparable around $\gamma t = 10.54$.

The survival probability fixes the yield available for observing that tail. In the same example, the surviving fraction is 1.18×10^{-15} at $\gamma t = 5$ and 2.02×10^{-25} at $\gamma t = 8$ [Fig. 1(c)]. The conditional rate describes the shape of the retained signal; a sensing assessment must also count the preparations lost before readout. The exact tail prediction therefore does not by itself establish a measurable sensing advantage.

System size and pump order. At fixed pump order $d > 1$, the large-system analysis predicts

$$\Delta_{\rm all} : \Gamma_D : \Gamma_N \longrightarrow 1 : 2 : \sqrt{2(d+1)}. \tag{2}$$

The imbalance rate approaches twice the unrestricted gap, and the particle-number signal is faster still. By contrast, single-particle pumping gives the exact ratio $1 : 2 : 2$ for nonzero transients: the two density rates coincide. Multiparticle pumping thus separates two physically allowed, parity-even signals, beyond the exclusion of odd modes.

The finite-size spectra in Fig. 2 support the semiclassical hierarchy. Its derivation requires nonzero observable overlaps and identification of the slowest radial modes with the local semiclassical levels. Existing spectral constraints [30] and numerical edge data through $n = 512$

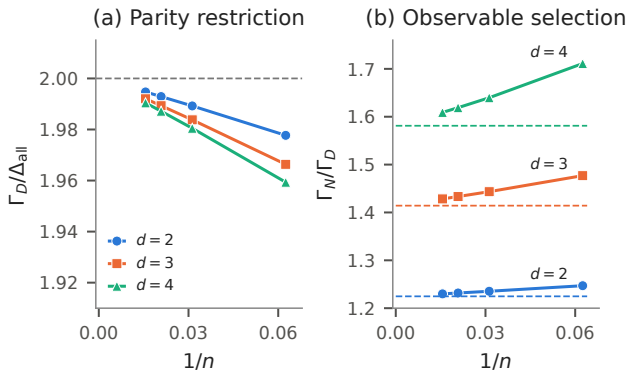


FIG. 2. Finite-size evidence for the predicted hierarchy, Eq. (2). (a) Imbalance decay divided by the unrestricted gap; in these cases it also equals the even-sector gap divided by the unrestricted gap. (b) The internal decay spacing of the number channel divided by imbalance decay. This is Γ_N/Γ_D when the particle-number overlap is nonzero. Symbols use $n = 16, 32, 48, 64$; dashed limits are 2 in (a) and $\sqrt{(d+1)/2}$ in (b). Ratios are independent of positive pump strength.

support that identification; a uniform global spectral remainder remains unproved. The leading edge for $d = 2$ recovers the known result of Ref. [31]. SM Secs. IV–V separate these asymptotic assumptions from the exact finite-size dynamics and give the derivation and prior-art comparison.

Within the monitored domain, the scaling $z = \zeta n^{1-d}$ with fixed $0 < \zeta < d!$ keeps the predicted density relaxation rates finite while the long-time survival decay rate grows in proportion to n. Conditional relaxation thus retains a finite timescale as the retained fraction decreases. This yield cost applies to observing the long-time tail; pump discrimination can use the earlier endpoint signal described below.

Detecting the pump with short trials. Let $a = z\binom{n}{d}/n \in [0, 1]$ denote the pump strength relative to the physical upper limit. We compare two known alternatives $a_0 < a_1$, with equal prior probabilities and all other model parameters known. Each preparation uses the same input and waiting time under either hypothesis. Its output is the retained system or one orthogonal failure flag f; every failed attempt counts toward the total number of preparations M.

A retained positive-number endpoint provides a short-time pump signal. The protocol starts from vacuum, waits $t = 1/(n\gamma)$, and measures total number or failure. Without pumping, such an occupied endpoint is impossible; at pump strength a, its probability is at least a/e. For the ideal zero-pump comparison, this gives no false positives and a sufficient preparation count proportional to $1/a$ at fixed target error. Detection therefore uses rare pumping events before the conditional relaxation tail develops. SM Sec. VI gives the complete probabilities and the comparison of two nonzero strengths.

A matching information bound limits how much any allowed endpoint measurement can improve this scaling. Using channel-simulation methods [32] and quantum-state distinguishability bounds [33], we obtain

$$M_{\rm opt} = \Theta\left[(\sqrt{a_1} - \sqrt{a_0})^{-2}\right] \tag{3}$$

for a fixed target error below one half, with a_1 bounded away from one. Here Θ means matching upper and lower scaling bounds, with constants independent of n and d. The upper bound is achieved by vacuum number readout. The lower bound allows any common input, a retained ancilla, any waiting time, and a joint measurement after independent preparations. It applies to the specified system-or-erasure output; neither emitted fields nor additional rejected quantum outputs are accessible. SM Sec. VII contains the proof and its assumptions.

For 0 versus a, or a versus $2a$, Eq. (3) gives the inverse-strength cost of rare-event counting. For two nearby strengths separated by δ at a fixed positive base value, it gives the usual inverse-square cost in δ. Neither behavior is set by the fractional spectral exponent. The pump jump amplitude is proportional to $\sqrt{a}$, so its zero-rate boundary also differs from the analytic-jump setting of Ref. [24].

The bounds quantify the preparation savings from using short trials in the example of Fig. 1. To distinguish the original pump from a 20% increase, the numerically optimized vacuum protocol has $\gamma t = 0.10652$ and a sufficient count of 7480 preparations for 5% average error. For the original $p = 0.75$ preparation at $\gamma t = 5$, a necessary-count bound is approximately 6.49×10^{15} for the same total-number-or-failure readout. The comparison uses a sufficient short-time count and a necessary long-time count, not exact optima; the latter bound is specific to that readout. Searches over 54 parameter pairs select vacuum without establishing its universal optimality or the optimal constant in Eq. (3).

Equation (3) counts complete preparations. The vacuum pump rate is $an\gamma$, and each trial prepares n modes. Reset overhead, detector backgrounds, timing resolution, and implementation of the uniform d-particle pump lie outside this budget. The resulting scaling describes preparation cost within the model; a hardware-level metrological comparison would also require those resources.

Discussion. The same-state comparison identifies readout symmetry, together with fermion parity, as a selector of observable relaxation near this EP. Changing detector weights exposes a different decay channel, while the survival probability determines the yield of its conditional tail. Short vacuum trials use the same generator to detect the pump before that low-yield regime. The matching preparation bounds make the distinction operational: access to a late-time decay and detection of a known perturbation have different resource requirements.

The exact reduction applies to uniform channels and number-conserving quadratic dynamics, with the large-system hierarchy subject to the stated semiclassical conditions. The detection bound uses the specified system-

or-erasure output and does not require an EP. Detecting a known pump and substituting its strength into a spectral formula remains model-dependent; independent certification of the fractional spectral law would require competing physical models and measurement uncertainties. Within these conditions, the signal reductions and resource bounds determine which decay a density detector selects and how the preparation cost of pump discrimination scales.

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